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Conceptual Modeling of Cyber-Physical Gaps in Air Traffic Control

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Abstract

On March 8, 2014, a Malaysia Airlines Boeing 777 carrying flight MH-370 from Kuala Lumpur to Beijing disappeared during its flight. In this paper, we employ a model-based approach to capture and study the chain of events leading to the disappearance of the aircraft. We show that the critical reason for the aircraft's disappearance was a cyber-physical gap (CPG). The CPG is the difference between the real state of a physical entity and its state as perceived by cybernetic agents. Failure to acknowledge the CPG's existence, capture it in system models, and incorporate suitable mitigation mechanisms into systems exposed to CPG, has resulted in the past and may result in the future in severe consequences. In the MH-370 case, the plane's position and adherence to its flight course were misperceived by the air traffic control, resulting in failure to track and find the plane, to this day. Our approach to CPG analysis has been previously shown to assist both designers and users handle emergent situations resulting from CPGs, such as the Three-Mile Island nuclear reactor's partial meltdown accident in 1979. We propose several mitigation mechanisms, whose integration with air traffic control systems could help prevent similar cases in the future.

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1. Introduction

On March 8, 2014, a Malaysia Airlines Boeing 777 carrying flight MH-370 from Kuala Lumpur to Beijing disappeared during its flight. Last contact by the pilot with the Ho-Chi-Min air traffic control (ATC), followed by

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confirmation of the flight plan for MH370, had not raised any special concerns about what was about to unfold[†]. While the reasons for the disappearance of the flight remain mysterious, the ATC's failure to detect and respond to the deviation of the aircraft from its designated course is the main cause for the fact that the modified course of the airplane could not be reconstructed, and that the exact location where the airplane hit the surface of the Indian Ocean remains unknown to this day. Four years later, the plane could not be recovered. Only debris found around the Indian Ocean is evidence that the plane is buried somewhere in its bottom. In May 2018 the search operation was terminated, after billions of dollars were spent without success in finding the lost aircraft.

The MH-370 disappearance was the first in a series of aviation catastrophes that spanned 2014 and 2015. It included Malaysia Airlines Flight MH-17 on July 17, 2014, from Amsterdam to Kuala Lumpur, which was shot down above Ukraine; Germanwings Flight 9525 on March 24, 2015, from Barcelona to Dusseldorf, which was taken down by its co-pilot in a suicidal act above the French Alps – possibly inspired by MH-370; and Metrojet Flight 9268 on October 31, 2015, from Sharm-a-Sheikh, Egypt, to Saint Petersburg, Russia, which was exploded in midair by an onboard bomb as an act of terror by Islamic Groups in the Sinai Peninsula. This series of events was a wakeup call for the International Civil Aviation Organization (ICAO) regarding ATC regulations.

One of the reasons for MH-370's disappearance could be the fact that ATC did not realize the airplane had gone off-course. The airplane's position was visible to ATCs of several countries, including Thailand, Malaysia, Vietnam, Indonesia, and Singapore. However, the fact that airplane was not on-course was probably unknown to the ATCs since the plane's designated course had not been provided to them in advance by the airline company. Failure to notice that the plane was not on-course could have allowed ATC to detect the situation as it was. Specifically, moments in which the plane took unusual turns from its original course and later during its perpetrated course, were imperative in the detection of anomalous or unusual behavior.

The fact that the actual route of the airplane in physical space did not match the planned and approved route as known to the ATC is a manifestation of the Cyber-Physical Gap (CPG) [1]. A cyber-physical system (CPS) resides and acts concurrently in cybernetic space and physical space [2], [3]. It follows that cybernetic modules in CPSs must possess information about the physical domain that allows the former to monitor, command, and control the latter. Therefore, entities in cyber-physical environments exist dually in cyberspace and physical space. This dual existence of a real-world embodiment and a virtual representation has been defined as the Cyber-Physical Duality (CPD). CPG is the difference between the state of the physical embodiment and its informational representation.

In this paper we study the relevance of CPGs to ATC, in the context of airliner positioning and tracking. We use the MH-370 story as a reference case, to demonstrate how failure of the ATC system to detect mismatches in flight course compliance might result in airplanes going out of observability and controllability. Shortly after the MH-370 disappearance, we hypothesized that a misconception of the MH-370 situation by ATC was the cause for the airplane's disappearance, due to a mismatch between the airplane's physical state and the information that ATC had [4]. This paper further extends that hypothesis and applies the CPG-Aware Modeling and Engineering (CPGAME) framework [1] to analyze the situation and incorporate mitigation and response mechanisms into the system.

The rest of this paper is organized as follows: Section 2 discusses related work on ATC and CPGAME. Section 3 includes model-based analysis of the ATC process, gradually enhanced to become CPG-aware. In section 4 we discuss the results of this study, conclude, and propose future research.

Nomenclature

ATC	Air Traffic Control
CPD	Cyber-Physical Duality
CPG	Cyber-Physical Gap
CPGAME	CPG-Aware Modeling and Engineering
FAA	Federal Aviation Administration
IATA	International Air Transport Association
ICAO	The International Civil Aviation Organization

[†] Audio Recording available here: <https://www.bbc.com/news/world-asia-44285241>

2. Related Work

2.1. Air Traffic Control

The disappearance of MH-370 was a turning point in civil aviation – both for aviation authorities and airline operators. ICAO established an airliner tracking concept of operations (CONOPS) [5]. ICAO also recommends that airline operators establish an aircraft tracking program [6] and equip airliners with flight tracking devices in order to know where each airline's planes are within 15-minute intervals [7]. In addition, the Aircraft Tracking Task Force (ATTF), led by the International Air Transport Association (IATA), analyzed potential flight tracking solutions based on available technologies. The ICAO Annex 11 [8] includes guidelines for air traffic control operators regarding the treatment of *strayed aircraft* and *unidentified aircraft*. These guidelines include establishing communication with the aircraft, determining its position, and alerting military, other ATS units, and other aircraft. It should be noted, though, that the guidelines do not include any interaction with airlines or attempts to obtain visual contact with the aircraft. The paradigm shift towards airliner tracking is only in its infancy, and multiple challenges must be met in order to drive global adoption and operationalization.

2.2. The Cyber-Physical Gap

CPG has been a primary concern in cybernetics [9]–[13]. The notion of systemic misconception has been addressed and applied in system safety [14]–[17], cyber security [18]–[20], counter-terrorism [21], [22], and microbiology [23], [24]. The CPG-Aware Modeling and Engineering framework (CPGAME) [1] relies on three modeling phases: a) naïve, b) fault-aware, and c) CPG-aware. In phase (a) the analyst is specifying the system using a nominal, naïve model, which does not account for faults and failures. This is often the case with legacy models. In phase (b) the analyst extends the model to account for possible disruptions, anomalies, exceptions, errors, faults, and failures in the process. In phase (c) the analyst further extends the model to relax the assumption that the information about the failures is available to the control system. The CPG-aware model includes several key functionalities that the cybernetic component must exhibit, including a) entity acquisition, b) representation management, and c) interaction analysis. In addition, CPGAME asserts that any interaction of a cybernetic agent with an external entity must rely on the perceived state of the entity, as held by the agent, rather than on the actual state of the entity. However, the entity's response is based on its actual state rather than on the state perceived by the agent.

CPGAME has been shown to significantly improve the capability to mitigate or respond to CPGs in complex systems. It can be applied during system design stages but also as part of operational systems audits. A system may be CPG-aware without officially-adopted CPG-aware modeling and design approach. This could be due to demonstrated failure in early prototypes or similar systems. However, an unstructured approach may result in greater gaps throughout the system's lifecycle.

2.3. Object-Process Methodology (OPM)

OPM is one of several leading, state-of-the-art methodologies for model-based systems engineering (MBSE), and a notable modeling language next to Unified Modeling Language (UML), Systems Modeling Language (SysML), Enhanced Functional Flow Block Diagrams (EFFBD), Simulink, State-charts, Integrated Definition Framework (IDEF), and others [25]–[27]. As opposed to some of the aforementioned languages, OPM is also a modeling framework and an official ISO standard (ISO 19450) for modeling ISO standards [28].

OPM uses a minimal universal vocabulary that includes the concepts *object*, *state*, *process*, and *relation*. Accordingly, OPM has a single diagram kind that accommodates all these concepts, thus capturing the functional, structural, and procedural aspects of the system in a unified manner. In addition to its graphical modality – the diagram set – OPM has a textual modality: a structured textual specification in Object-Process Language (OPL), an English-like language, which includes an OPL paragraph for each diagram and a textual OPL sentence for each diagram construct. The relations among OPM diagrams (OPDs) include a) unfolding objects or processes (collectively defined as *things*) for structural or functional decomposition; b) zooming into objects and processes for

spatial or temporal ordering; and c) generating dedicate views for important things. These abstraction-refinement complexity management mechanisms are applicable at any depth level.

OPCAT is a freely available desktop modeling and simulation software application (which can be downloaded for free from <http://esml.iem.technion.ac.il/>) [29]. OPCAT supports most of the OPM ISO-approved notation. It automatically validates model edits according to OPM grammar, and automatically generates OPL text for each OPD, for each graphical construct. OPCAT provides model execution and simulation capabilities, which support model validation and verification.

The primary reasons for adopting OPM as an expression vehicle for CPD-related aspects in complex systems include: a) OPM's simple-yet-robust concept vocabulary, b) OPM's inherent support for and distinction of physical or informatical essence and environmental or systemic affiliation for things in a model, and c) The capability to evolve and validate the model gradually based on hierarchical decomposition and model simulation.

2.4. CPG-Aware Modeling and Engineering (CPGAME)

The modeling, simulation, analysis, and mitigation of CPG on top of nominal conceptual models of complex systems has been significantly enhanced due to the use of Object-Process Methodology, OPM [30]–[32]. Studies applying OPM for analyzing CPGs have found deficiencies and proposed mitigation mechanisms in areas as nuclear energy [1], airport services [4], [34], [35], cyber-security [33], medical robotics [36], and systems biology [37]. In [1], a CPG-aware analysis of the Three Mile Island nuclear reactor was conducted, accounting for the reasons for the partial meltdown accident of 28 March 1979 (TMI2), which was the severest US commercial nuclear power plant accident. TMI2 is clearly an outcome of CPG: a simple mechanical failure in a relief valve was not understood in real-time by the reactor control team. Due to that gap, the operators made the wrong decisions and took the wrong actions, exacerbated the situation, and led to partial meltdown [38].

CPGAME captures six types of CPGs as part of the main system model – rather than as a separate model or safety annex. This ensures direct treatment of CPG-related risks – and indeed other risks – within the functional specification of the system. The six types of CPG are listed in Table 1, with possible examples for ATC.

Table 1. Classes and Types of Cyber-Physical Gaps [1]

Class	Type	ATC Example
Class A: failure to detect or identify a detectable entity	Type A1: detection mechanism problem or lack of a pattern that resembles the entity	Failure to notice an airplane when it enters the ATC unit's airspace
	Type A2: acquisition mechanism problem or inability to perform entity pattern-matching	Failure to identify an object detected by the ATC as a civilian airplane
Class B: failure to generate a coherent representation of an acquired entity	Type B1: problem with representation of the acquired entity's state	Failure to notice that an airplane is not following its designated route
	Type B2: problem with representation of the acquired entity's behavior	Failure to understand that a rogue aircraft is not complying with ATC instructions
Class C: failure to generate a coherent interaction with an acquired entity	Type C1: problem with generating the intended interaction	Failure to communicate an instruction to a pilot
	Type C2: problem with generating the intended result of an interaction	Failure to obtain pilot compliance with an ATC instruction

In addition, CPGAME evaluates models by checking each couple of systemic agent and external entity for compliance with five criteria: a) the specification of an internal representation, b) specification of an acquisition process, c) specification of a representation process, d) specification on actions and interactions in a representation-based (rather than actual state-based) manner, and e) specification of interaction analysis.

3. Conceptual Modeling of ATC Protocol

In this section we model the ATC protocol. We conform to CPGAME and go through three modeling phases: naïve, fault-aware, and CPG-aware. We begin with the naïve model of the ATC process, once an airliner enters the ATC's airspace. The initial model is naïve because it assumes and demonstrates a completely nominal and expected behavior on both sides of the Airplane-ATC interaction. The naïve model can be built by or rely on inputs from

subject-matter experts. It also serves as a good starting point for those who are new to model-based analysis. This model is shown in Fig. 1. It includes an OPM diagram and its corresponding textual OPL specification. The model is in undergoing simulation, which enables the analyst to validate the model. When the simulated model was captured, the Airplane's two processes – **Flying** and **Communicating with ATC** – were occurring. On the ATC's side, the **Route Communicating** process was also occurring, generating a new Route Instruction to the **Airplane**, which is acknowledged by the pilots, as indicated by the red token along the output link from the process **Communicating with ATC** towards the object **Route Instruction Acknowledgment**. According to this model, whenever an **Airplane's Location** is *in-Airspace*, it triggers **Airplane Tracking** and subsequent ATC procedures.

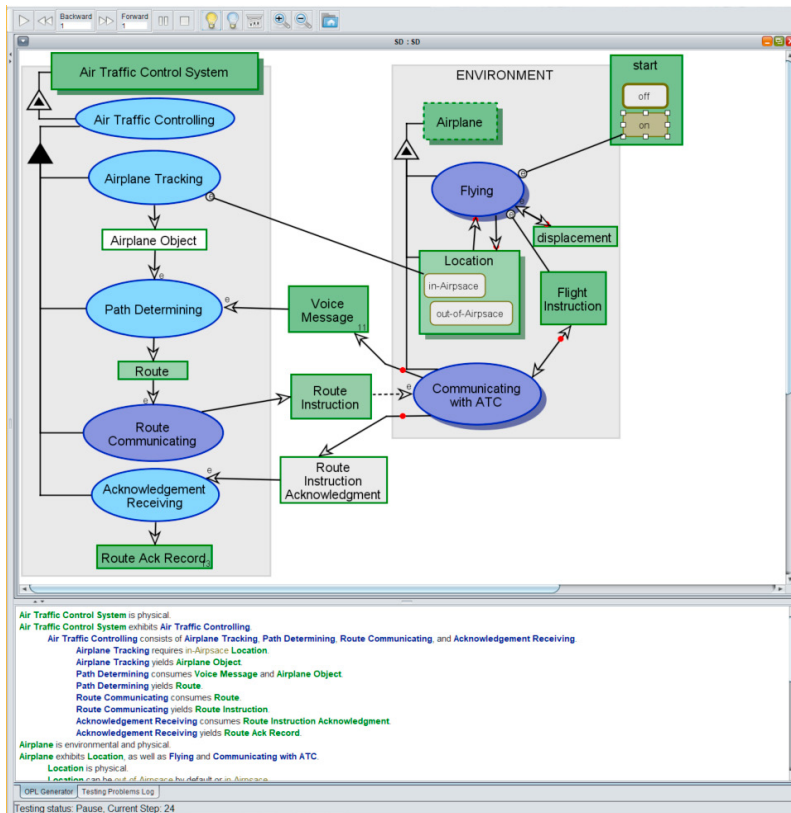


Fig. 1. Naïve Model of the Air Traffic Control system, showing the nominal process of airline tracking and route authorization

The next phase in the process is the elaboration of the naïve model into a fault-aware model, which accounts for possible faults and failures in the process. A revised fault-aware model is shown in Fig. 2. Possible fault and failure modes are specified as negative states of objects. For instance, any message sent over radio communication between the ATC and Airplane might be either received or lost. When a message is lost, its loss triggers another attempt to communicate, until it succeeds. In addition, when a voice message from the pilot is lost, i.e., does not arrive at ATC when expected, it also triggers another control cycle by the ATC. Another fault is related to the compliance of the pilot or airplane with the instruction that it receives. Although this model now covers several failure modes and exceptional situations, it is still not CPG-aware. Any information is assumed to be available to both parties. In Fig. 2, which was captured during a simulation of the model, we see that the current lack of compliance by the airplane is triggering Airplane Tracking, i.e., another control cycle. Similarly, a lost Voice Message from the pilots also reactivates this process. At the same time, it seems that the latest Route Instruction was lost, but the pilot did send a Route Instruction Ack on the previous Route Instruction. In this arbitrary, randomized case, we can conclude that unstable radio communication between ATC and the Airplane is preventing the Airplane from receiving new instructions, resulting in non-compliance of the pilots with ATC's instructions.

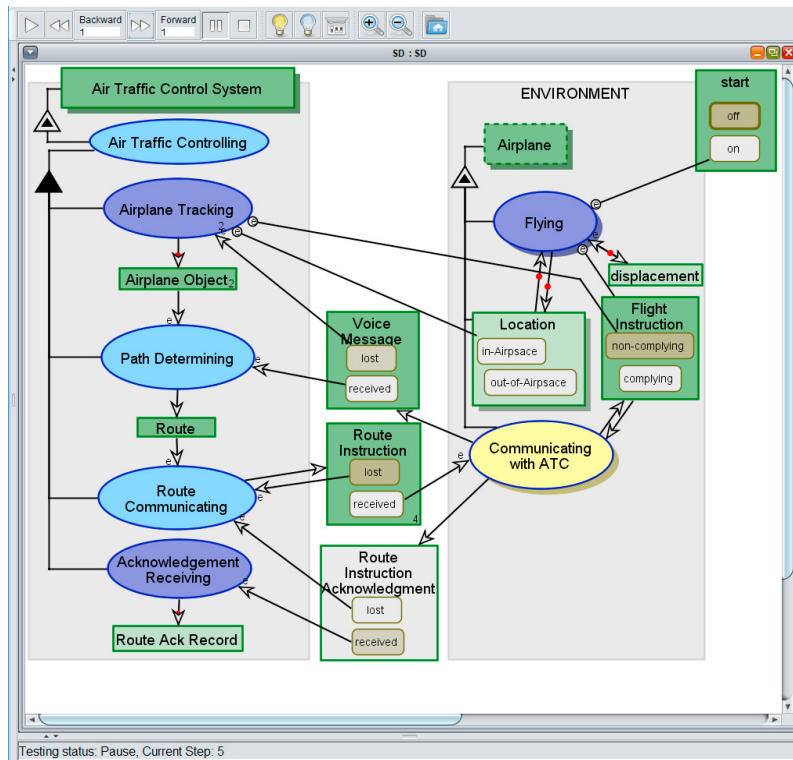


Fig. 2. Fault-Aware Model of the Air Traffic Control system

The third phase of the process is the transformation of the model into a CPG-aware one. There are many potential CPGs in this process, but due to the limited scope of the present paper, we will focus on the most critical issue, which applies to the ATC's awareness of the airplane's intended route and its determined route. The ATC should acquire information from Peer ATCs and from the Airline Company, in order to have better affinity between its route decision and the airplane's current bearing. In addition, the ATC must detect whether an Airplane is entering airspace or leaving airspace, and communicate with Peer ATCs either to a) obtain Inbound Route Information or b) Handoff the Airplane to the next ATC and provide it with Outbound Route Information. This extension of the model, as illustrated in Fig. 3, reflects the emerging operational concept that extends the boundaries of the ATC protocol beyond the direct point-to-point interaction between the pilot and the ATC operator. In Fig. 3, it can be seen that following several cycles of simulation, the ATC has acquired information about the Airplane from a Peer ATC and from the Airline Company. At the moment the simulation was captured, the ATC was conducting a Handoff of an Airplane leaving the airspace to the next ATC in line. Since in this case the Airplane was non-complying with latest Route Instructions, both ATCs, the Airline Company and possibly military forces should ensure that the Airplane is not a strayed Airplane or that it is not flown by a pilot with malicious intents. These extensions are beyond the current scope, but the potential for extending the model further are clear.

4. Discussion and Conclusion

In this paper, we applied model-based analysis of cyber-physical gaps in air traffic control, notably in the context of correct and continuous airplane route tracking, acquisition, determination, and compliance monitoring. Following the Cyber-Physical Gap-Aware Modeling and Engineering (CPGAME) framework, we gradually extended a naïve model of interaction between an airplane and an ATC unit towards a fault-aware model that accounts for various issues such as communication disruptions, and then towards a CPG-aware model that accounts for gaps in the

information and knowledge owned by the ATC unit regarding airplanes entering its controlled airspace. Updates to the ATC protocol in the aftermath of MH-370's disappearance and new guidelines by ICAO and IATA were reflected in the CPG-aware model, especially regarding interactions with peer ATCs and with the Airline Company regarding the whereabouts and flight plans of entering airplanes. This procedure can dramatically decrease chances of losing track or control over commercial airplanes. Additional regulations and guidelines can be validated and analyzed using our model-based approach.

Due to the limited scope of this paper, we could not elaborate the internal logic of the control processes taking place in the ATC. We believe that further analysis and modeling of these procedures will uncover additional issues and discrepancies that require additional regulatory and technical revisions to the ATC protocol. The current model is focusing on a single ATC-airplane interaction. A continuous model escorting the airplane from take-off to landing can discover potential disruptions and discrepancies in the process and further enhance the protocol. Future work should focus on encouraging commercial aviation regulation and standardization organizations to adopt a formal approach to the specification of procedures and contingencies in air traffic management.

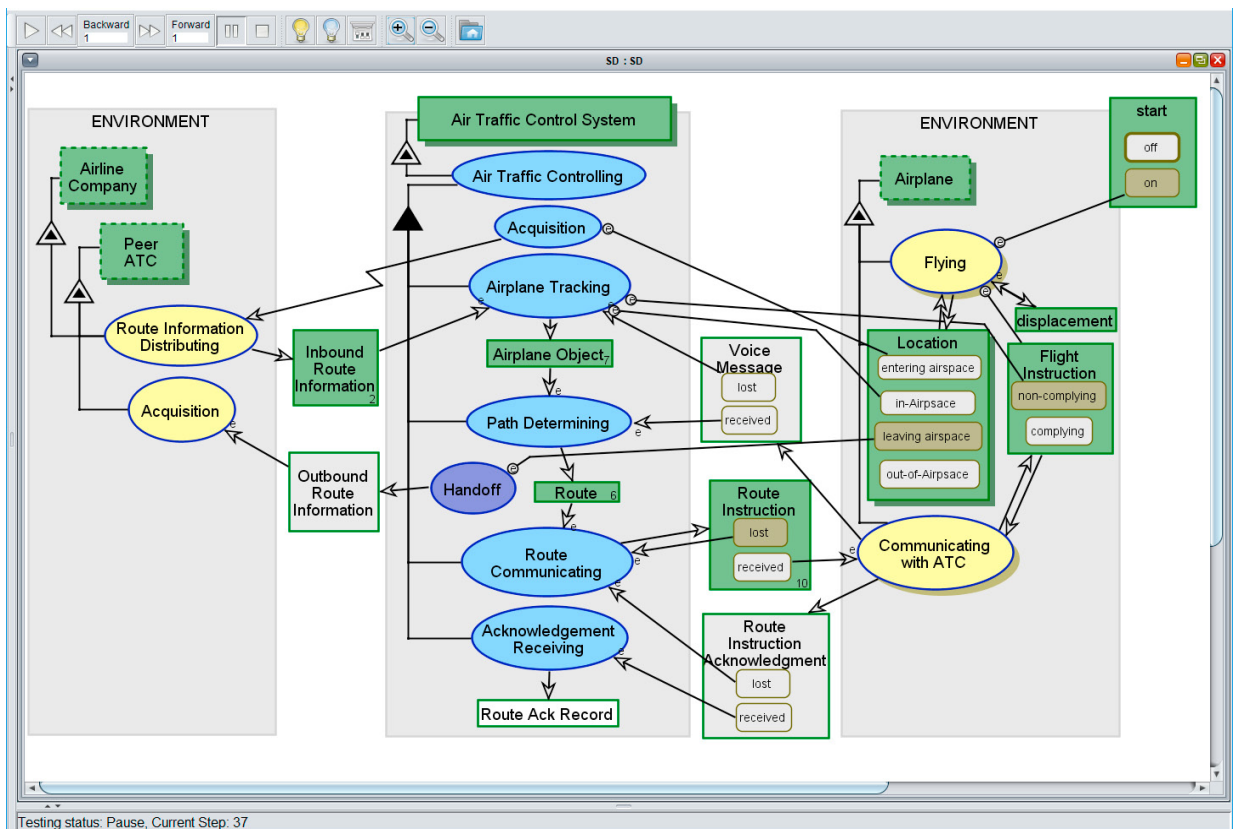


Fig. 3. CPG-Aware Model of the Air Traffic Control system

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